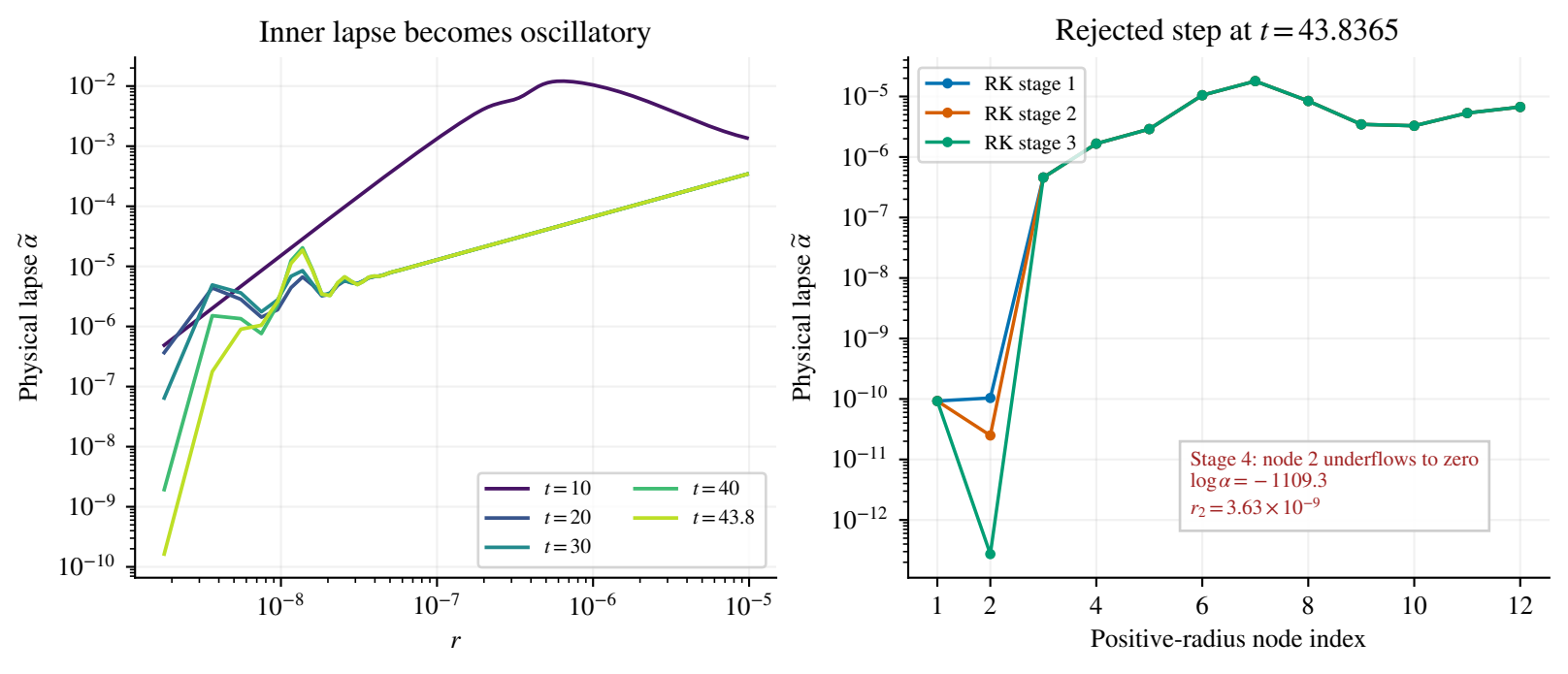
Inner lapse becomes oscillatory
Physical lapse $\tilde{\alpha}$
r
$t = 10$
$t = 20$
$t = 30$
$t = 40$
$t = 43.8$
Rejected step at $t = 43.8365$
RK stage 1
RK stage 2
RK stage 3
Stage 4: node 2 underflows to zero
$\log \alpha = -1109.3$
$r_2 = 3.63 \times 10^{-9}$
Positive-radius node index